

# One Plot, Two Reporting Strategies

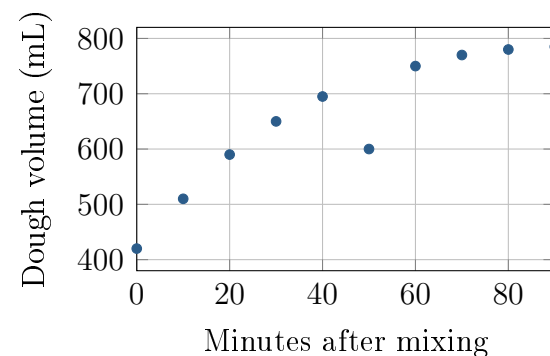
Unit 5 · Topic 5.1 · THE PROBLEM

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

**Jules:** Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

**Ravi:** Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.

Selected measurements (minutes, mL): (0, 420), (20, 590), (70, 770), (90, 785).



## FIRST TAKE — one sentence before you work the parts

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.

**DOK 1** (a) Name the explanatory and response variables, with units.

**DOK 2** (b) Find the volume increase from 0 to 20 minutes and from 70 to 90 minutes. Is the 50-minute batch above or below the nearby batches?

**DOK 3** (c) Whose reporting strategy is supported? Describe the direction, form, strength, and unusual point in context, using your two increases. Explain how a straight-line summary could mislead someone predicting later growth. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.